

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final
Duration: 2 Hours

Semester :Spring 2024
Full Marks: 40

CSE 422: Artificial Intelligence

Answer the following questions.
Figures in the right margin indicate marks.

Name: _____	ID: 2. _____	Section: 01
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1. CO4

SL no.	unique_ID	Age	Income	Student?	Credit Rating	Buys_Computer?
1	#5	30	High	NO	Fair	NO
2	#19	30	High	NO	Excellent	NO
3	#24	32	High	NO	Fair	YES
4	#2	41	Medium	NO	Fair	YES
5	#10	42	Low	YES	Fair	YES
6	#32	44	Low	YES	Excellent	NO
7	#4	37	Low	YES	Excellent	YES
8	#8	28	Medium	NO	Fair	NO
9	#6	31	Low	YES	Fair	YES
10	#7	36	Medium	YES	Fair	YES
11	#9	22	Medium	YES	Excellent	YES
12	#14	35	Medium	NO	Excellent	YES
13	#13	37	High	YES	Fair	YES
14	#21	42	Medium	NO	Excellent	NO

- Among the features “Income” and “Student”, which one do you think would be more suited for classifying if a person would buy a computer or not? **Provide** answer using ID3 4
- Do you think “unique_ID” would be a good feature for classification here? **Provide** a brief explanation to support your claim 2
- Can we use “Age” as a building node for the decision tree? If not, briefly **explain** what should be done beforehand. 2
- For row number 4,10 and 11 if the value of “Buys_Computer” becomes “Soon”, then **compute** the Information Gain for feature “Credit Rating”. 2

2. CO2

CARB University (CARBU) wants to start a new degree program which is B.A. in Social Media Influencing (SMI). Suppose the degree program is associated with the following courses:

SMI101: Fundamentals of Social Media Influencing	SMI251: Increasing Engagements
SMI111: Principles of Shorts	SMI360: Finance, Advertisement and Boosting
SMI422: Representation and Problem-Solving	SMI321: TikTok
SMI421: Media Networking	SMI320: Instagram
SMI220: Reels	

In order to graduate from the degree program, one must complete the following four requirements:

Requirement 1: (SMI101 AND SMI111) OR (SMI101 AND SMI422) OR (SMI421 AND SMI251)

Requirement 2: SMI422 OR SMI421 OR SMI220

Requirement 3: SMI251 OR SMI321 OR SMI360

Requirement 4: SMI422 OR SMI320

In addition, the department imposes the following restrictions:

Restriction 1: Students can take only one class from the set SMI422, SMI421, and SMI320.

Restriction 2: Students can't take both SMI101 and SMI360.

Restriction 3: Students can't take both SMI251 and SMI321.

In addition, courses cannot be used to count towards multiple graduation requirements - so if you use

Set A

SMI422 to fulfill part of Requirement 1 it can't count towards either Requirement 2 or Requirement 4. Answer the following,

- a. X just started his junior year at CARBU, and needs to graduate as soon as possible. Suppose all he has left to take are SMI required classes. **Determine** the initial domains and at least 4 constraints of his trying to find a set of classes to satisfy all requirements as a CSP (Hint: the requirements should be variables). 5
- b. Suppose X has already taken SMI320 towards his Requirement 4 and SMI101 towards Requirement 1. **Determine** other classes he must take to graduate using backtracking and forward checking and explain how forward checking helps. 5

3. CO3 a. A patient went to the hospital for a malaria test. The doctors informed him that their test can successfully diagnose malaria positive given the patient is actually malaria positive 94% of the time. Also, the probability of having no malaria and getting a malaria positive test result is 3%. Meanwhile, 29% of people in general who come for tests are malaria positive. Now if the patient is already diagnosed malaria positive, then **calculate** the probability of the patient actually being malaria negative. 6
- b. Is being malaria positive and having a negative test result independent of each other? **Show** calculation and figure out using the information from question A. (Keep two digits after the decimal point of the final score) 4

4.CO4

House ID	Size of House (sq ft)	Selling Price (USD)
1	1200	150,000
2	1500	185,000
3	1800	220,000
4	2100	260,000
5	2400	300,000

Let size of the houses be the input variable (also known as predictor), x and the selling prices of the houses be the output variable (also known as response), y . To predict the value of y , we establish a linear relationship $h_w(x) = w_1x + w_0$.

- a. **Interpret** the meaning of the weights, w_1 and w_0 in terms of linear regression 1
- b. Let us define the Loss-function, $SSR(h_w) = \sum_{j=1}^N (y_j - (w_1x_j + w_0))^2$ and the [partial-derivative of 4.5

this Loss-function with respect to each weight is given by $\frac{d(SSR(h_w))}{dw_0} = -2(y - h_w(x))$ and $\frac{d(SSR(h_w))}{dw_1} = -2x(y - h_w(x))$.

Tahia uses her special calculation and estimates the weights as $w_1 = 1.5$ and $w_0 = 0$. Using Gradient-descent algorithm up to 2 iterations, improve these estimates. You may consider the value of learning parameter $\alpha = 1$. **Show** all steps of calculation.

No. of iteration	$h(w)$	$Loss(h_w)$	$\frac{d(Loss(h_w))}{dw_0}$	$\frac{d(Loss(h_w))}{dw_1}$	w_0	w_1

- c. Using the updated estimates of weights w_0 and w_1 from question 2, calculate the predicted selling price for a 2100 sq ft house. Then, **compare** this prediction with the actual selling price from the dataset to evaluate the accuracy of the model. 1.5
- d. **Explain** why it is important for the derivative of the loss function with respect to each weight to be (near) zero for achieving better predictions in a model. 1.5
- e. **Discuss** the impact on the gradient descent algorithm when the learning rate increases over time. 1.5